Parallel, Distributed, and Quantum Exact Single-Source Shortest Paths with Negative Edge Weights

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Negative Weight Shortest Path

- Find s, t shortest path in a directed weighted graph or report a negative cycle.
- Weights are possibly negative, polynomially bounded integers.

- Notations:
  - number of vertices n,
  - number of edges m,
  - $\tilde{O}(f) = O(f \cdot \mathrm{polylog}(f))$.

Parallel Computing

- Work: total amount of unit operations.
- Depth: the length of the longest chain of computations.

Previous Works

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Our work	$m^{1+o(1)}$	$n^{1/2+o(1)}$

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Our work	$n^{o(1)}$ calls to non-negative SSSP	

Further Implications

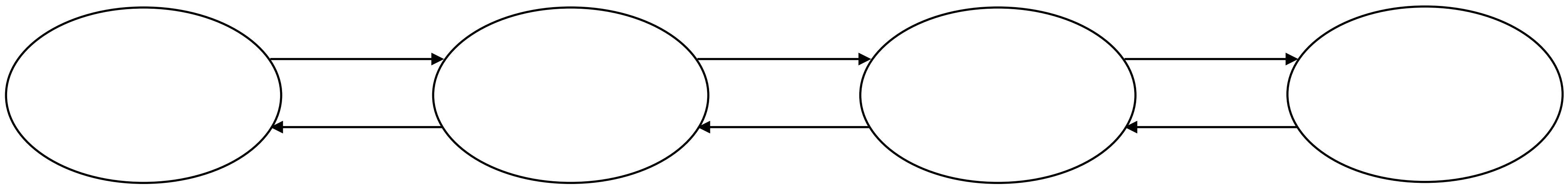
- Our reduction works on other models: distributed, streaming, quantum, etc.
  - PRAM: $m^{1+o(1)}$ work, $n^{1/2}$ depth
  - CONGEST: $\left(n^{\frac{2}{5}}D^{\frac{2}{5}} + \sqrt{n} + D\right) \cdot n^{o(1)}$
  - Streaming: $\tilde{O}(n)$ space, $n^{\frac{1}{2}+o(1)}$ passes
  - Quantum query:
    - $n^{1.5+o(1)}$ queries to adjacency matrix or
    - $(mn)^{0.5+o(1)}$ queries to adjacency list

Techniques

- [Bernstein, Nanongkai, Wulff-Nilsen'FOCS22] can be viewed as a reduction to $n^{o(1)}$ non-negative weight SSSP given directed low diameter decomposition (LDD).
- Our contribution: directed LDD with $\tilde{O}(1)$ calls to non-negative SSSP.

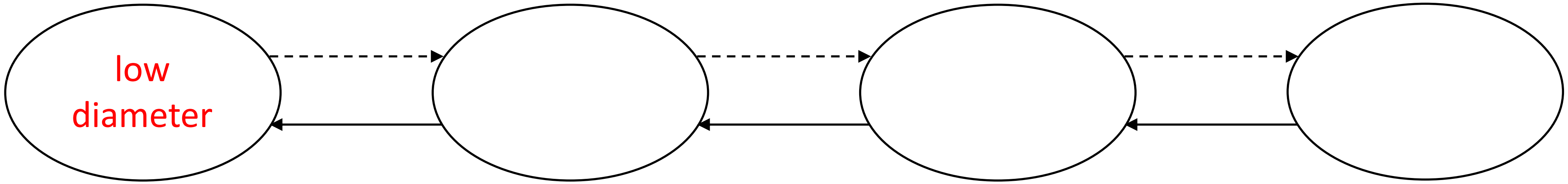
Low Diameter Decomposition

- Delete a set of edges and call each of the remaining strongly connected components *clusters*



Low Diameter Decomposition

- Delete a set of edges and call each of the remaining strongly connected components *clusters*
  - Each cluster has weak low diameter (diameter $\tilde{O}(d)$)
  - Each edge is deleted with small probability ($\tilde{O}(1/d)$)



Undirected Graph

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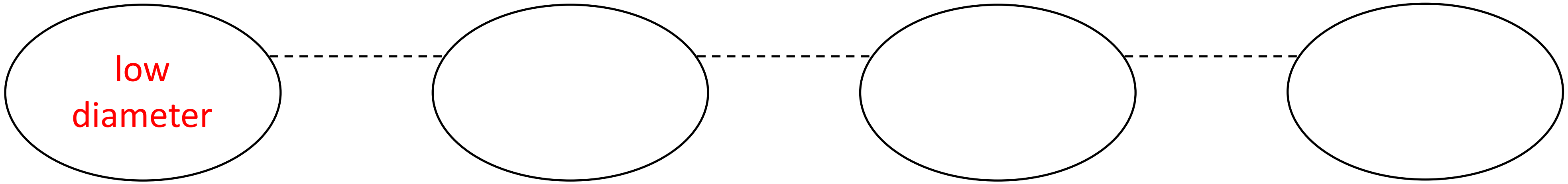
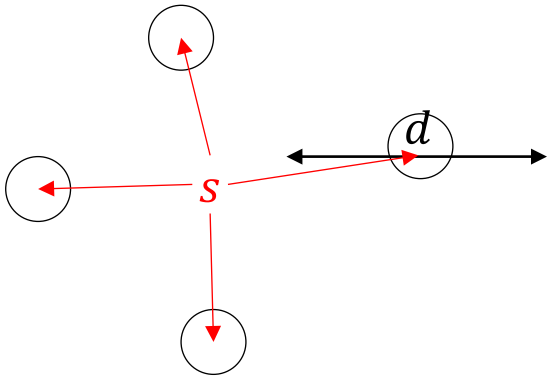


Undirected Graph

- Undirected graph: [Miller, Peng, Xu'SPAA13] $\tilde{O}(m)$ work and $\tilde{O}(d)$ depth.

- Idea: random growing non-overlaping balls

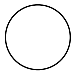
*Can be achieved by one non-negative SSSP call: add dummy source s with edges pointing to centers of clusters, run SSSP from s



- Does not work for directed graph: a ball is not necessarily strongly connected

Directed Graph

- Sequential algorithm:
  - Grow a ball with random radius



Directed Graph

- Sequential algorithm:
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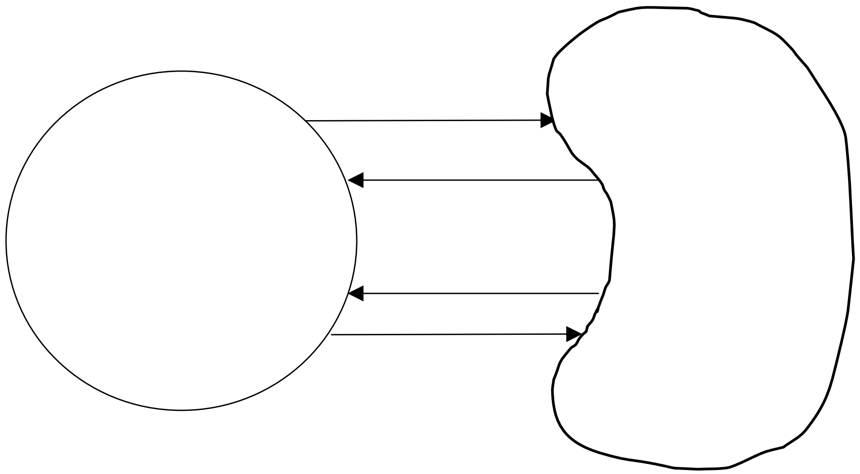
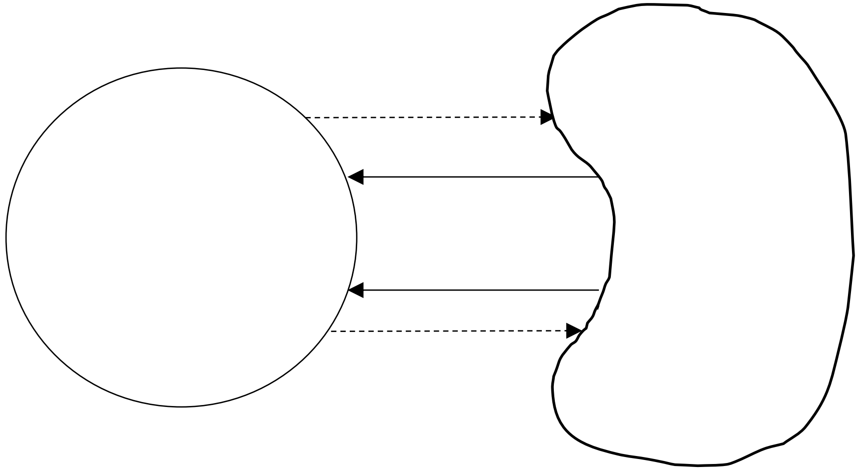
  - Cut outgoing edges (spit the graph into two parts)



Directed Graph

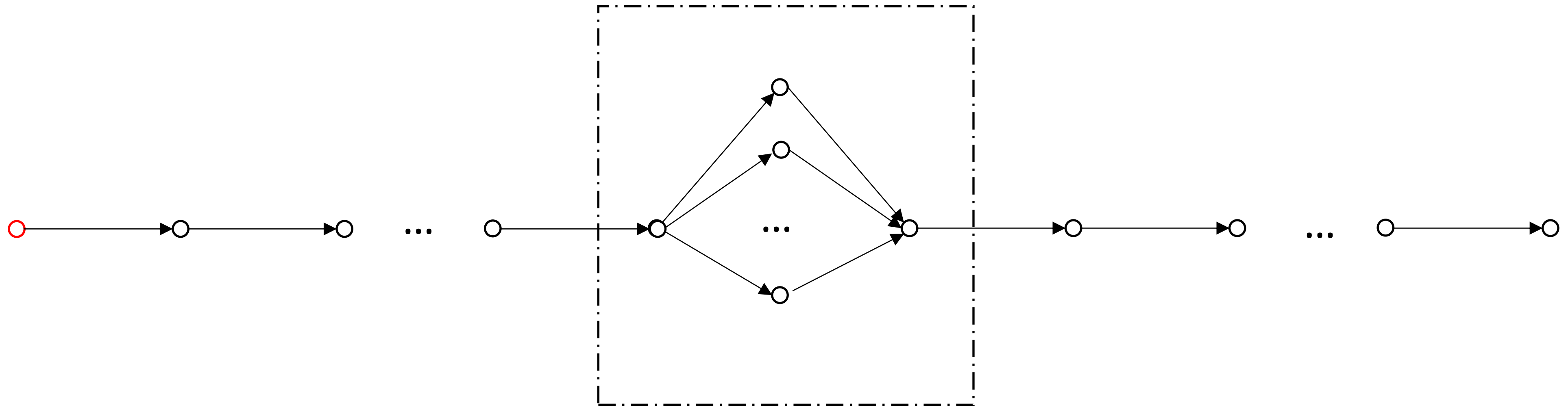
- Sequential algorithm:
  - Grow a ball with random radius
  - Cut outgoing edges (spit the graph into two parts)
  - Recurse on both side until the whole graph has diameter d (becomes a cluster)



- High depths if the ball has $O(1)$ vertices.
- Can we make the recursion balanced, i.e., grow a ball with $\approx n/2$ vertices?

Directed Graph (Low Depth)

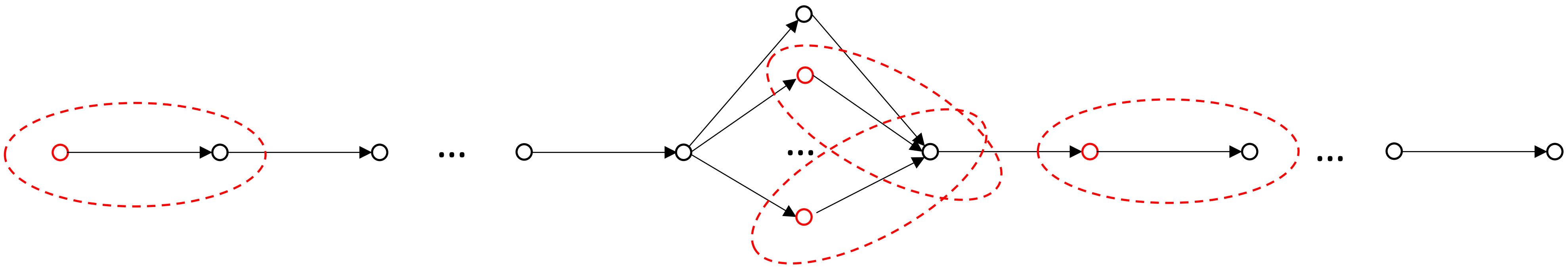
- Can we make the recursion balanced, i.e., grow a ball with $\approx n/2$ vertices?
  - Not possible if we grow a ball from one vertex



Most nodes are here

Directed Graph (Low Depth)

- Can we make the recursion balanced, i.e., grow a ball with $\approx n/2$ vertices?
  - Our approach: growth balls from multiple nodes
  - Recurse on the union of the balls



- How many nodes to pick?
  - Use binary search to find the right threshold

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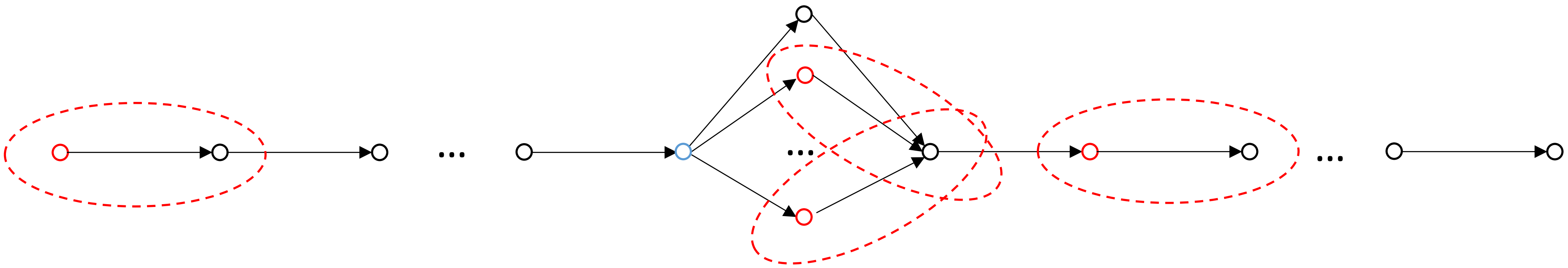

- How many nodes to pick?
  - Use binary search to find the right threshold
  - Avoid vertices with large outgoing ball

Summary

- Reduction from (possibly) negative weight SSSP to non-negative weight SSSP
  - With $n^{o(1)}$ calls
  - In PRAM, CONGEST, quantum, streaming, …
- Directed low diameter decomposition by $\tilde{O}(1)$ calls to non-negative weight SSSP